

tain a common standard to exchange data with government for automation.

Similarly, Matter allows a device vendor to write everything in a particular format, which allows other devices to talk to that device or a cloud platform to provide additional services. Imagine a security system telling a light bulb, as per Matter specifications, to turn on when it detects motion, so that camera can capture the intruder's picture. Here all the devices such as motion sensor, bulb, and camera could be from different vendors, but they talk to each other observing the Matter standard.

Whenever a computer mouse is connected to the central processing unit (CPU) of a computer, data goes from the mouse to the computer in a very structured way. To enhance security, Samsung conducted different experiments of their own hardware and learnt from them. From these learnings and participating companies' feedback, Samsung has introduced additional security in Matter.

Earlier scenario

Earlier, a small manufacturer in the Internet of Things (IoT) smart home ecosystem had to work with multiple big players and pay each of them, even if a small amount. There were many small players who could not manage multiple stock keeping units (SKUs).

With Matter, we now have one certification which will be honoured by all the other players like Samsung Hub, Google Nest, Alexa, and Apple. This drastically reduces the cost of certifying a device for a small manufacturer and makes it easier for him to sell the same product in multiple markets and verticals.

Sometimes some of the certifications have region-specific limitations. For example, some of the products that are certified in the United States may not work in India. Matter is a great logo to have for most device

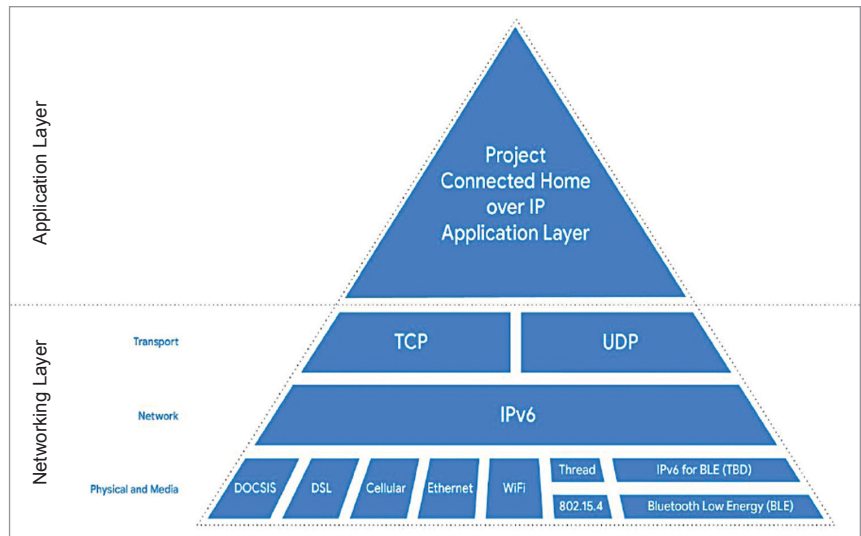


Fig. 1: Protocol stack

manufactures. System integrators, especially the small players, who want to use off-the-shelf products and deploy IoT services will greatly benefit.

Earlier there could be instances where a company's strength lied purely in hardware manufacturing or a service on top of that. To maintain certifications, government compliance, and user data privacy security, the company needed the full stack—from trusted hardware platform to transmission, cloud platform, and apps—to be secure and securely exchanging data.

Matter has a strict regime for security; Samsung devices provide additional Knox hardware security. Samsung puts five to six cameras in some of their phones and it is not difficult for them to use them as a monitoring device. But their biggest concern has always been privacy and security.

We have all heard of incidents where smart door locks were used to launch denial of service attacks (DDOS) on public servers where smart locks started bombarding the servers. For example, if a banking server gets attacked with thousands of locks sending requests to the server, it would deny access to their regular customers wanting banking

services. With IoT devices expected to multiply, the attack potential also becomes exponentially large.

Security systems usually encrypt messages between devices with two software keys. It is like a bank locker where one key is owned by the bank and the other by the client. These messages can only be decrypted with the right keys.

One of the causes of security breaches in devices is the rooting of a device, where it switches off the binary system of a device and the security keys are duplicated. Samsung provides Knox Guard to deal with this. Knox Guard provides clustered hardware platforms, which means the keys are burnt during manufacturing and get stored in a secured zone where it cannot be overwritten by any software switch.

United States Air Force and their defence establishment has certified the Knox grade platform for mobility devices, and they use it for their pilots' tablets, etc. The same Knox Guard is used in Samsung washing machines, televisions, and other smart appliances.

Matter has defined a concept for multiple admins. It means that you can onboard a device to one application (for example, SmartThings app)